

Bihar Engineering University, Patna

B.Tech. 3rd Semester Examination, 2023

Course: B.Tech.

Code: 104303

Subject: Mathematics-III

Time: 03 Hours

Full Marks: 70

Instructions:-

- (i) The marks are indicated in the right-hand margin.
- (ii) There are **NINE** questions in this paper.
- (iii) Attempt **FIVE** questions in all.
- (iv) Question No. 1 is compulsory.

Q.1 Choose the correct option/ fill in the blank of the following (Answer any seven): [2 x 7 = 14]

- (a) The value of Chebyshev polynomials $T_2(x)$ is
 - (i) $2x^2 - 1$
 - (ii) $2x^2 + 1$
 - (iii) $4x^3 + 3x$
 - (iv) $4x^3 - 3x$
- (b) If R is a relation on a finite set having n elements, then the number of relations on A is
 - (i) 2^n
 - (ii) 2^{n^2}
 - (iii) n^2
 - (iv) n^n
- (c) Let R be a relation on a set A such that $R = R^{-1}$, then R is
 - (i) reflexive
 - (ii) transitive
 - (iii) symmetry
 - (iv) None of the above
- (d) Let $f(x) = \frac{ax+b}{cx+d}$ then $f \circ f(x) = x$ provided
 - (i) $d = -a$
 - (ii) $d = a$
 - (iii) $a = b = c = d = 1$
 - (iv) $a = b = 1$
- (e) The maximum number of edges in a simple graph with n vertices is
 - (i) n
 - (ii) $\frac{n-1}{2}$
 - (iii) $\frac{n(n-1)}{2}$
 - (iv) None of these
- (f) SD is defined as
 - (i) $\sqrt{\frac{\sum f(x-\bar{x})^2}{\sum f}}$
 - (ii) $\frac{\sum f(x-\bar{x})}{\sum f}$
 - (iii) $\frac{\sum f(x-\bar{x})^2}{\sum f}$
 - (iv) None of the above
- (g) Let A and B are two possible outcomes of an experiment and suppose $P(A) = 0.3$, $P(B) = K$, $P(A \cup B) = 0.6$. If A and B are mutually exclusive events then the value of K is
 - (i) 0.1
 - (ii) 0.2
 - (iii) 0.3
 - (iv) 0.4
- (h) If the mean of poisson distribution is m , then SD of this distribution is
 - (i) m^2
 - (ii) $\sqrt{m}$
 - (iii) m
 - (iv) None of the above
- (i) The median of the numbers 11, 10, 12, 13, 9 is
 - (i) 12.5
 - (ii) 12
 - (iii) 10.5
 - (iv) 11
- (j) A hypothesis is false but is accepted, then there is an error of type _____

Q.2 (a) Find the generating function of Chebyshev polynomials [7]

(b) Show that $T_{n+1}(x) = 2xT_n(x) - T_{n-1}(x)$, where $T_n(x)$ is Chebyshev polynomials. [7]

Q.3 (a) What is the wavelet transform? [7]

(b) For any three sets A, B, C prove that $A \times (B \cup C) = (A \times B) \cup (A \times C)$. [7]

- Q.4 (a) Prove that the relation R on the set $N \times N$ defined by $(a, b)R(c, d) \in R \Leftrightarrow a+d = b+c$ for all $(a, b), (c, d) \in N \times N$ is an equivalence relation. [7]
- (b) Show that the function $f: Q \rightarrow Q$ given by $f(x) = 2x - 3$ for all $x \in Q$ is a bijection. [7]

- Q.5 (a) Show that : [7]

$$\int_0^p x(\operatorname{ber}^2 x + \operatorname{bei}^2 x) dx = p(\operatorname{ber} p \operatorname{bei}' - \operatorname{bei} p \operatorname{ber}' p)$$

- (b) Consider the following graph: [7]



Write the vertex set and the edge set, and give a table showing the edge-endpoint function.

- Q.6 (a) Discuss Skewness and Kurtosis for the following frequency distribution : [7]

Marks	0-10	10-20	20-30	30-40	40-50
No. of student	5	10	40	20	25

- (b) In a partially destroyed laboratory record of an analysis of a correlation data, the following results only are eligible: [7]

Variance of $x = 9$

Regression equations:

$$40x - 18y = 214, 8x - 10y + 66 = 0$$

Find (i) mean values of x and y , (ii) the standard deviation of y and coefficient of correlation between x and y , and (iii) angle between the lines of regressions.

- Q.7 (a) Find the mean and variance of binomial distribution. <https://www.akubihar.com> [7]

- (b) The probability that a pen manufactured by a company will be defective is $1/10$. If 12 such pens are manufactured, find the probability that (i) exactly two will be defective, (ii) at least two will be defective and (iii) none will be defective. [7]

- Q.8 (a) In a test 2000 electric bulbs, it was found that the life of a particular make was normally distributed with an average life of 2040 hours and SD of 60 hours. Estimate the number of bulbs likely to burn for (i) more than 2150 hours, (ii) less than 1950 hours and (iii) more than 1950 hours and less than 2160 hours. [7]

- (b) Find the curve of best fit of the type $y = ae^{bx}$ to the following data by method of least square: [7]

X	1	5	7	9	12
Y	10	15	12	15	21

- Q.9 (a) The mean of a certain normal population is equal to the standard error is equal to the standard error of the mean of the samples of 100 from that distribution. Find the probability that the mean of the sample of 25 from the distribution will be negative. [7]

- (b) An unbiased coin is thrown n times. It is desired that the relative frequency of the appearance of head should lie between 0.49 and 0.51. Find the smallest value of n that will ensure this result with 90% confidence. [7]

Bihar Engineering University, Patna

B.Tech 3rd Semester Examination, 2024

Course: B.Tech
Code: 100311

Subject: Mathematics-III (Differential Calculus)

Time: 03 Hours
Full Marks: 70

Instructions:-

- (i) The marks are indicated in the right-hand margin.
- (ii) There are NINE questions in this paper.
- (iii) Attempt FIVE questions in all.
- (iv) Question No. 1 is compulsory.

Q.1 Choose the correct option / answer the following (Any seven question only):

[2 x 7 = 14]

(a) The solution of the differential equation $y'' - 3y' + 2y = e^{2x}$

- (i) $C_1 e^x + C_2 e^{2x} + \frac{e^{2x}}{2}$
- (ii) $C_1 e^{-x} + C_2 e^{-2x} + \frac{e^{2x}}{2}$
- (iii) $C_1 e^{-x} + C_2 e^{2x} + \frac{e^{2x}}{2}$
- (iv) None of these

(b) The solution of $p + q = z$ is

- (i) $f(xy, y \log z) = 0$
- (ii) $f(x + y, y + \log z) = 0$
- (iii) $f(x - y, y - \log z) = 0$
- (iv) None of these

(c) The series $1 - \frac{1}{(1!)^2} \left(\frac{x}{2}\right)^2 + \frac{1}{(2!)^2} \left(\frac{x}{2}\right)^4 - \frac{1}{(3!)^2} \left(\frac{x}{2}\right)^6 + \dots$ is

- (i) $J_{1/2}(x)$
- (ii) $x J_{1/2}(x)$
- (iii) $x J_0(x)$
- (iv) None of these

(d) If $\int_{-1}^1 P_n(x) dx = 2$, then n is

- (i) 0
- (ii) 1
- (iii) 2
- (iv) None of these

(e) The particular solution of $y'' + 9y' = \sin 3x$ is

- (i) $-\frac{x}{6} \cos 3x$
- (ii) $\frac{x}{6} \cos 3x$
- (iii) $-\frac{3x}{2} \cos 3x$
- (iv) $\frac{3x}{2} \cos 3x$

(f) The differential equation $y'' + (x^3 \sin x)^5 y' + y = \cos x^3$ is

- (i) Homogeneous
- (ii) Non-linear
- (iii) Second order linear
- (iv) Non-homogeneous with constant coefficient

(g) The magnitude of the directional derivative of the function $f(x, y) = x^2 + 3y^2$ in a direction normal to the circle $x^2 + y^2 = 2$, at the point (1, 1) is

- (i) $4\sqrt{2}$
- (ii) $5\sqrt{2}$
- (iii) $7\sqrt{2}$
- (iv) $3\sqrt{2}$

(h) If $\vec{R} = x\vec{i} + y\vec{j} + z\vec{k}$ is the position vector, then $\text{div } \vec{R}$ is

- (i) 1
- (ii) 3
- (iii) -3
- (iv) -2

(i) The degree of the homogeneous function $f(x, y) = \frac{x^2 + y^2 + z^2}{x+y}$ is

- (i) 4/5
- (ii) 4/3
- (iii) 7/3
- (iv) None of these

(i) If $f(x, y) = \tan^{-1} y/x + \sin^{-1} x/y$, then $x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} =$

(i) 2

(ii) 1

(iii) 0

(iv) -1

Q.2 (a) If $y^{1/m} + y^{-1/m} = 2x$, prove that [7]

(i) $(x^2 - 1)y_2 + xy_1 - m^2y = 0$

(ii) $(x^2 - 1)y_{n+2} + (2n + 1)xy_{n+1} + (n^2 - m^2)y_n = 0$

(b) If $p^2 = a^2 \cos^2 \theta + b^2 \sin^2 \theta$, prove that $p + \frac{d^2 p}{d\theta^2} = \frac{a^2 b^2}{p^3}$ [7]

Q.3 (a) Use Taylor's theorem to expand $f(x, y) = x^y$ in powers of $(x - 1)$ and $(y - 1)$. [7]

(b) Find the dimension of the rectangular box, open at the top, of maximum capacity whose surface area is 432 sq.cm. [7]

Q.4 (a) If $u = \tan^{-1} \left(\frac{y^2}{x} \right)$, prove that $x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = -\sin^2 u \cdot \sin 2u$ [7]

(b) If by the substitution $u = x^2 - y^2, v = 2xy, f(x, y) = \theta(u, v)$, show that $\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = 4(x^2 + y^2) \left(\frac{\partial^2 \theta}{\partial u^2} + \frac{\partial^2 \theta}{\partial v^2} \right)$ [7]

Q.5 (a) Verify Stoke's theorem for $F = (2x - y)I - yz^2J - y^2zK$ over the upper half surface $x^2 + y^2 + z^2 = 1$, bounded by its projection on the xy -plane. [7]

(b) If the directional derivative of $\phi = ax^2y + by^2z + cz^2$ at the point $(1, 1, 1)$ has maximum magnitude 15 in the direction parallel to the line $\frac{x-1}{2} = \frac{y-3}{-2} = z$, find the values of a, b, c [7]

Q.6 (a) Solve $(x^2 - y^2)(xdx + ydy) = a^2(xdy - ydx)$ [7]

(b) Solve $p(p + x) = y(x + y)$ [7]

Q.7 (a) Solve $(2y \log y + y - x)dy = ydx$ [7]

(b) Solve, by the method of variation of parameters, $y'' - 2y' + y = e^x \log x$. [7]

Q.8 (a) Solve $x^2 \frac{d^2 y}{dx^2} - 3x \frac{dy}{dx} + y = \log x \frac{\sin(\log x) + 1}{x}$ [7]

(b) Prove that $\int_m(x) = \frac{1}{\pi} \int_0^\pi \cos(m\theta - x \sin \theta) d\theta$, where 'n' is positive integer [7]

Q.9 (a) Solve the equation: $(z^2 - 2yz - y^2)p + (xy + zx)q = xv - zx$. [7]

(b) Solve the equation: $(x - y)(px - qy) = (p - q)^2$ [7]

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Instructions:-

- (i) The marks are indicated in the right-hand margin.
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- (iv) Question No. 1 is compulsory.

Q.1 Choose the correct answer of the following (Any seven question only):

[2 x 7 = 14]

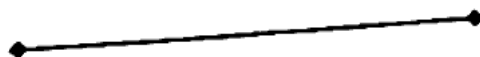
- (a) The value of $\lim_{x \rightarrow 0} \left(\frac{\sin x}{x}\right)^{1/x}$ is
 - (i) 0
 - (ii) 1
 - (iii) e
 - (iv) 1/e
- (b) The value of the integral $\int_C \{yzdx + (xz + 1)dy + xydz\}$ Where C is any path from (1, 0, 0) to (2, 1, 4) is
 - (i) 6
 - (ii) 7
 - (iii) 8
 - (iv) 9
- (c) The maximum value of $\sin x + \cos x$ is
 - (i) 1
 - (ii) 2
 - (iii) $\sqrt{2}$
 - (iv) 0
- (d) The value of $\nabla^2 [(1-x)(1-2x)]$ is equal to
 - (i) 2
 - (ii) 3
 - (iii) 4
 - (iv) 6
- (e) The degree of the differential equation $y \frac{dx}{dy} - \left(\frac{dx}{dy}\right)^2 - \sin y \left(\frac{dx}{dy}\right)^3 - \cos x = 0$ is
 - (i) 0
 - (ii) 1
 - (iii) 2
 - (iv) Cannot be determined
- (f) If $z = \tan^{-1} \frac{y}{x}$, then $\text{div}(\text{grad } f)$ is equal to
 - (i) 1
 - (ii) -1
 - (iii) 0
 - (iv) 2
- (g) If P_n is the Legendre polynomial of first kind, then the value of $\int_{-1}^1 x P_n P'_n dx$ is
 - (i) $\frac{2}{(2n+1)}$
 - (ii) $\frac{2n}{(2n+1)}$
 - (iii) $\frac{2}{(2n+3)}$
 - (iv) $\frac{2n}{(2n+3)}$
- (h) If J_n is the Bessel's function of first kind, then the value of $\int_{-\frac{1}{2}}^{\frac{1}{2}}$ is
 - (i) $\sqrt{\frac{2}{\pi x}} \left(\frac{\cos x}{x} - \sin x\right)$
 - (ii) $\sqrt{\frac{2}{\pi x}} \left(\frac{\sin x}{x} - \cos x\right)$
 - (iii) $\sqrt{\frac{2}{\pi x}} \sin x$
 - (iv) $\sqrt{\frac{2}{\pi x}} \cos x$
- (i) The solution of $p \tan x + q \tan y = \tan z$ is
 - (i) $\sin x / \sin y = \varphi(\sin y / \sin z)$
 - (ii) $\sin x \cdot \sin y = \varphi(\sin y / \sin z)$
 - (iii) $\sin x / \sin y = \varphi(\sin y, \sin z)$
 - (iv) $\sin x / \sin y = \varphi(\sin y \cdot \sin z)$
- (j) The vector $\vec{v} = e^x \sin y \hat{i} + e^x \cos y \hat{j}$ is
 - (i) Solenoidal
 - (ii) irrational
 - (iii) rotational
 - (iv) cannot be found

- Q.2** (a) Form the partial differential equation $(x-a)^2 + (y-b)^2 + z^2 = 1$. [7]
 (b) Solve $xp + yq = 3z$. [7]
- Q.3** (a) Find the directional derivative of $\phi = z^2yz + 4xz^2$ at the point $(1, -2, 1)$ in the direction of the vector $2\hat{i} - \hat{j} - 2\hat{k}$. [7]
 (b) Find a unit vector normal to the surface $x^3 + y^3 + 3xyz = 3$ at the point $(1, 2, -1)$. [7]
- Q.4** Solve the following questions:-
 (a) Solve partial differential equation $\frac{y^2z}{x}p + xzq = y^2$. [7]
 (b) Show that the function $f(x, y) = \begin{cases} \frac{xy}{\sqrt{x^2+y^2}}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$ [7]
 is continuous at origin.
- Q.5** (a) If $f = (x^2 + y^2 + z^2)^{-n}$, then find $\text{div grad } f$ and determine n , if $\text{div grad } f = 0$. [7]
 (b) Verify Green's theorem for $\int_C \{(xy + y^2)dx + x^2dy\}$ [7]
 Where C is bounded by $y = x$, $y = x^2$.
- Q.6** (a) Evaluate the integral by changing the order of integration [7]

$$\int_0^{\infty} \int_0^x xe^{-\frac{x^2}{y}} dy dx$$

 (b) Solve the differential equation [7]
 $(x^2 + y^2 + x) dx - (2x^2 + 2y^2 - y) dy = 0$
- Q.7** Verify the stokes' theorem for [14]
 $A = (y - z + 2)\hat{i} + (yz + 4)\hat{j} - xz\hat{k}$
 Where S is the surface of the cube $x = 0, y = 0, z = 0, x = 2, y = 2$ and $z = 2$ above the xy -plane. <https://www.akubihar.com>
- Q.8** (a) Prove that [6]
 $2nJ_n(x) = x(J_{n-1}(x) + J_{n+1}(x))$ [8]
 (b) Prove that

$$\sum_{n=0}^{\infty} \frac{x^{n+1}}{n+1} P_n(1) = \frac{1}{2} \log \left(\frac{1+x}{1-x} \right)$$
- Q.9** Solve the following questions: [7]
 (a) Using Green's theorem, evaluate $\int_C [(y - \sin x) dx + \cos x dy]$ where C is the [7]
 plane triangle enclosed by the lines $y = 0, x = \frac{\pi}{2}$ and $y = \frac{2x}{\pi}$
 (b) Prove that $\text{div}(r^n \vec{r}) = (n+3)r^n$. Hence show that $\text{div} \left(\frac{\vec{r}}{r^3} \right)$ is solenoidal.



Bihar Engineering University, Patna

B.Tech. 3rd Semester Examination, 2023

Course: B.Tech.

Code: 100312

Subject: Mathematics III (PDF, Probability & Statistics)

Time: 03 Hours

Full Marks: 70

Instructions:-

- The marks are indicated in the right-hand margin.
- There are NINE questions in this paper.
- Attempt FIVE questions in all.
- Question No. 1 is compulsory.

Q.1 Choose the correct answer of the following (any seven question only):

[2 x 7 = 14]

- The order of the PDE obtained by eliminating f from $Z = f(x^2 + y^2)$ is
(i) First (ii) Second (iii) Third (iv) Fourth
- Who was the first person to develop the heat equation?
(i) Galileo Galilei (ii) Joseph Fourier
(iii) Daniel Gabriel (iv) none of these
- The linear partial differential equation of order one is known as
(i) Lagrange's linear PDE (ii) Cauchy's linear PDE
(iii) Charpit's linear PDE (iv) none of the above
- Poisson distribution is
(i) Discrete (ii) Continuous (iii) Normal (iv) None of these
- Two unbiased coins are tossed simultaneously. The probability of getting less than 3 tails is
(i) 0 (ii) 1 (iii) 3/4 (iv) 1/2
- The total area under the normal curve above the x-axis is
(i) 0.5 (ii) 1 (iii) 2 (iv) 1.5
- The median for the series: 4, 6, 9, 4, 2, 8, 10 is
(i) 4 (ii) 7 (iii) 6 (iv) 8
- The relation between mean, median and mode is
(i) Mode = 3Median - 2 Mean (ii) Mode = 2Median - 3 Mean
(iii) Mean = 3 Median - Mode (iv) none of these
- If the regression coefficients are 0.8 & 0.2, what would be the value of coefficient of correlation?
(i) 0.6 (ii) 0.5 (iii) 0.4 (iv) 0.3
- The expectation of the number of heads in 15 tosses of a coin is
(i) 15 (ii) 30 (iii) 7.5 (iv) 8.5

Q.2 (a) Form partial differential equation by eliminating the function f from $Z = e^{ax+by} f(ax - by)$. [7]

(b) Solve the linear partial differential equation [7]

$$\frac{\partial^2 z}{\partial x^2} + 2 \frac{\partial^2 z}{\partial x \partial y} + \frac{\partial^2 z}{\partial y^2} = \sin(2x + 3y).$$

Q.3 (a) Obtain the general solution of heat flow equation $k \left(\frac{\partial^2 u}{\partial x^2} \right) = \frac{\partial u}{\partial t}$ by the method of separation of variables. [7]

(b) Show that a general solution of wave equation $c^2 \left(\frac{\partial^2 \phi}{\partial x^2} \right) = \frac{\partial^2 \phi}{\partial t^2}$ is $\phi = f(x + ct) + g(x - ct)$. [7]

Q.4 (a) Obtain solution of Laplace's Equation in cylindrical polar coordinates. [7]

(b) Determine whether the following equation is hyperbolic, parabolic, and elliptic? [7]

$$x \frac{\partial^2 u}{\partial t^2} + t \frac{\partial^2 u}{\partial x \partial t} + \frac{\partial^2 u}{\partial t^2} = 0.$$

- Q.5 (a) If X is a random variable such that $E[X] = 3$ and $E[X^2] = 13$, use the Chebyshev's inequality to determine the lower bound for $P[-2 < X < 8]$. [7]
- (b) Two integers are selected at random from 1 to 11. If the sum is even, find the probability that both the numbers are odd. [7]

- Q.6 (a) State and prove Baye's theorem. [10]
- (b) Find the mean number of heads in three tosses of a coin. [4]

- Q.7 (a) The first four moments of a distribution about $x = 2$ are 1, 2.5, 5.5 and 16. Calculate the first four moments about the mean. [7]
- (b) The following table gives the number of aircraft accidents that occurred during the various days of the week. Find whether the accidents are uniformly distributed over the week. (The tabulated value of Chi-square at 5% level for 6 degree of freedom is 12.59).

Days	Sun.	Mon.	Tue.	Wed.	Thu	Fri	Sat	Total
No. of accidents	14	16	8	12	11	9	14	84

- Q.8 (a) A random sample of 5 college students is selected and their grades in Mathematics & Statistics are found to be: [7]

Mathematics	85	60	73	40	90
Statistics	93	75	65	50	80

- Calculate Spearman's rank coefficient.
- (b) By the method of least square, find the straight line that best fits the following data: [7]

x	1	2	3	4	5
y	14	27	40	55	68

- Q.9 Write short notes on any two of the following: [7x2=14]

- (a) Binomial distribution
- (b) Poisson Distribution
- (c) Normal Distribution

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B.Tech 3rd Semester Exam., 2021

(New Course)

MATHEMATICS—III

(PDE, Probability and Statistics)

Time : 3 hours

Full Marks : 70

Instructions :

- (i) The marks are indicated in the right-hand margin.
- (ii) There are **NINE** questions in this paper.
- (iii) Attempt **FIVE** questions in all.
- (iv) Question No. 1 is compulsory.
- (v) Statistical data is given at the end.
- (vi) Non-programmable scientific calculator is allowed.

1. Choose the correct answer (any seven) :

$2 \times 7 = 14$

(a) If $z = f\left(\frac{xy}{z}\right)$ satisfies the partial differential equation $x^\alpha p - yq = \beta$, where $p = \frac{\partial z}{\partial x}$ and $q = \frac{\partial z}{\partial y}$, then the value of

 (α, β) is

- (i) (1, 0)
- (ii) (0, 1)
- (iii) (0, 0)
- (iv) (1, 1)

(b) The particular integral of the PDE

$$xy \frac{\partial^2 z}{\partial x \partial y} = 1$$

is $\log(ax)\log(by)$. Then the value of (a, b) is

- (i) (1, 1)
- (ii) (2, 1)
- (iii) (2, 3)
- (iv) (1, 2)

(c) The solution of the PDE

$$y^2 zp + x^2 zq = y^2 x$$

is $x^\alpha - y^\beta = f(x^2 - z^2)$. Then the value of (α, β) is

- (i) (2, 3)
- (ii) (3, 3)
- (iii) (3, 1)
- (iv) (1, 1)

(d) Let A and B are two possible outcomes of an experiment and suppose $P(A) = 0.3$, $P(B) = \frac{K}{7}$ and $P(A \cup B) = 0.6$.

If A and B are independent events, then the value of K is

- (i) 1
- (ii) 2
- (iii) 3
- (iv) 4

- (e) The first three moments of a distribution about 5 are 2, 7, 15 respectively. Then the mean and variance of the distribution are

(i) (5, 15)

(ii) (7, 15)

(iii) (7, 52)

(iv) (7, 3)

- (f) If two dice are thrown and the probability that the sum is neither 8 nor 12 is $\frac{5}{k}$, then the value of k is

(i) 3

(ii) 4

(iii) 6

(iv) 12

- (g) A random variable X has mean 2 and variance 3. If the upper bound for $P(|X-2| > 6)$ by using Chebyshev's inequality is $\frac{1}{K}$, then the value of K is

(i) 9

(ii) 12

(iii) 15

(iv) 18

- (h) If the mean of exponential distribution

$$f(x) = \begin{cases} ke^{-kx}, & x > 0 \\ 0, & x \leq 0 \end{cases}$$

is $\frac{1}{6}$, then the value of k is

(i) 4

(ii) 5

(iii) 6

(iv) 7

- (i) Let X and Y have joint p.d.f.

$$f(x, y) = \begin{cases} cxy; & 0 < x < 2, 0 < y < 1 \\ 0; & \text{elsewhere} \end{cases}$$

and $E(xy) = \frac{p}{q}$. Then the value of (p, q) is

(i) (4, 9)

(ii) (4, 4)

(iii) (8, 9)

(iv) (9, 4)

- (j) Two random variables have the least square regression lines with equations $2x + y = 5$ and $3x + 2y = 8$. Then the correlation coefficient between two variables x and y is

(i) $\frac{\sqrt{3}}{2}$

(ii) $-\frac{\sqrt{3}}{2}$

(iii) $\frac{1}{2}$

(iv) $-\frac{1}{2}$

2. (a) Obtain the partial differential equation by eliminating the arbitrary function

$$u = f(x^2 + y^2 + z^2, z^2 - 2xy)$$

- (b) Solve

$$z^2 (p^2 + q^2) = x^2 + y^2$$

where $p = \frac{\partial z}{\partial x}$ and $q = \frac{\partial z}{\partial y}$.

7+7

3. (a) Solve the partial differential equation $(2D^2 - DD' - 3D'^2)z = 5e^{x-y}$.

- (b) Prove that

$$(1 - x^2)P'_n(x) = n[P_{n-1}(x) - xP_n(x)]$$

where $P_n(x)$ is the Legendre's polynomial of the first kind.

7+7

4. Two ends A and B of a rod of length 20 cm have the temperatures 30°C and 80°C , respectively, until the steady-state conditions prevail. Then the temperatures at the ends A and B are changed to 40°C and 60°C , respectively. Find the temperature distribution of the rod at any time t .

14

5. (a) Let A, B and C be three independent events. Then show that B and $A \cap \bar{C}$ are also independent.

- (b) A coin is tossed. If it turns up H, two balls will be drawn from urn A, otherwise 2 balls will be drawn from urn B. Urn A contains 3 red and 5 blue balls, urn B contains 7 red and 5 blue balls. What is the probability that urn A is used? It is given that both balls are blue.

(Find in both cases, when balls were chosen with replacement and without replacement.)

7+7

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(Turn Over)

6. (a) If X and Y are independent Poisson variates such that $P(X=1)=P(X=2)$ and $P(Y=2)=P(Y=3)$. Find the variance of $X-2Y$.

- (b) Show that the recurrence relation between the moments of binomial distribution is given by

$$\mu_{r+1} = \left(nr\mu_{r-1} + \frac{d\mu_r}{dp} \right) pq$$

where μ_r is the r th order moment about the mean.

7+7

7. Given the joint probability density function of X and Y as

$$f(x, y) = \begin{cases} kxy; & 0 < x < 4, 0 < y < 3 \\ 0; & \text{elsewhere} \end{cases}$$

- (a) Compute the probability density function of $X+2Y$.

- (b) Compute the following :

(i) $P(1 < y \leq 2)$

(ii) $P(1 < x+2y \leq 3)$

(iii) $P(x > 1/y \leq 2)$

8+6

8. The table gives the stopping distance y (feet) of an automobile travelling at speed x (miles per hour) as the instant danger is sighted :

x	30	45	60	75	90	105
y	16	27	40	60	87	115

- (a) Fit a least-square parabola of the form $y = a + bx + cx^2$ to the data.

- (b) Estimate y when $x = 60$ miles per hour and 120 miles per hour.

Also compute the coefficient of correlation for the above data.

14

9. It has been found from experience that the mean breaking strength of a particular brand of thread is 275.6 grams with a standard deviation of 39.7 grams. Recently a sample of 36 pieces of thread showed a mean breaking strength of 253.2 grams. Can one conclude at a significance level of (a) 0.05 and (b) 0.01 that the thread has become inferior?

7+7

Given that $f(t) = \frac{1}{\sqrt{2\pi}} \int_0^t e^{-\frac{x^2}{2}} dx$, then

$$f(2.0) = 0.4772, f(2.41) = 0.4920, f(1.58) = 0.4429,$$

$$f(0.47) = 0.1808, f(1.645) = 0.45, f(1.96) = 0.4750,$$

$$f(0.41) = 0.1591, f(1.28) = 0.40, f(1.23) = 0.39,$$

$$f(1.475) = 0.43, f(2.33) = 0.49, f(2.58) = 0.495$$

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